

MindNav: Knowledge-Graph Tool Calling for Training-Free Multi-Robot Object Navigation

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Abstract

Multi-robot object navigation at matched 7B parameter scale exposes a sharp divide between perception–reasoning interfaces: VLM-based systems that look at raw images (MCo-CoNav, EfficientNav) collapse to 0% success, while map-based systems that serialize the scene to text (Co-NavGPT) retain viable performance. We argue the interface, not the model capacity, is the binding constraint: a 7B VLM asked to reason spatially from raw images underperforms a 7B text LLM given a clean symbolic summary. We instantiate this view as **MindNav**, a training-free multi-robot navigator that shares the Co-NavGPT perception stack (Detectron2 + RedNet + FMM) but replaces flat-text frontier assignment with LLM tool-calling over a dynamically constructed knowledge graph of rooms, objects, doors, and robots. On HM3D VAL_MINI with $N=2$ robots and Qwen2.5-7B-Instruct, MindNav reaches SR=0.733 and SPL=0.416, improving SPL by +36% relative over Co-NavGPT’s text-LLM frontier assignment at identical perception and LLM scale.

1 Introduction

LLM-driven multi-robot object navigation has converged on two interface designs for the perception–reasoning boundary. Co-NavGPT (Yu et al., 2023) serializes a shared 2D semantic map into room–object text and hands it to a central text LLM that assigns frontiers to robots. MCo-CoNav (Gao et al., 2025) replaces the text LLM with a VLM that reads raw images and panoramic stitches. Both pipelines share the same Detectron2+RedNet perception stack and differ mainly in what the reasoner consumes.

At matched 7B parameter scale, these two choices lead to very different outcomes. In our reproductions on HM3D VAL_MINI, Co-NavGPT’s text-LLM assignment reaches SR=0.700, while

MCoCoNav and EfficientNav’s VLM-based assignment collapse to SR=0.000 despite using the same exploration infrastructure. A “random frontier” ablation reaches SR=0.760, already above every VLM variant, pinning the collapse on the reasoner rather than the exploration loop.

We read this as evidence that *the interface between perception and reasoning is the binding constraint* at 7B scale, not the model family. A VLM asked to reason about spatial relationships from raw image tokens is competing with itself, using visual attention to both detect and reason. A text LLM given a clean symbolic summary reasons from a better-prepared state.

We take the argument one step further. Co-NavGPT’s flat room–object text discards structure: it does not carry certainty aggregated across detections, room connectivity, or the spatial diversity between frontiers. We therefore propose **MindNav**, a training-free multi-robot navigator that shares Co-NavGPT’s perception stack but replaces flat text with a dynamically constructed knowledge graph of four node types (Room, Object, Robot, Door) and six edge types (containment, proximity, room connectivity, etc.). The central LLM calls tools over this KG: estimate $P(\text{target} \mid \text{room})$ from parametric world knowledge, then assign frontiers with a diversity-aware objective.

Contributions.

1. A **perception–reasoning interface diagnostic** on HM3D val_mini at matched 7B scale: VLM (0%) vs text LLM (70%) vs random (76%), isolating the interface as the binding constraint.
2. **MindNav**, a KG tool-calling frontier-assignment module that replaces Co-NavGPT’s text substrate, with incremental KG construction, noisy-OR certainty aggregation, and spatial-diversity-aware frontier assignment.
3. An **empirical result**: SR=0.733, SPL=0.416

on HM3D val_mini with $N=2$ robots and Qwen2.5-7B, a +36% relative SPL improvement over Co-NavGPT’s text-LLM frontier assignment at matched infrastructure and model scale.

2 Related Work

2.1 LLM / VLM Multi-Robot ObjectNav

Co-NavGPT (Yu et al., 2023) established the central-LLM frontier-assignment recipe with Detectron2+RedNet perception, a 2D semantic map, and flat-text LLM reasoning. MCoCoNav (Gao et al., 2025) swaps the text LLM for a VLM on the same substrate; EfficientNav (Anonymous, 2025) explores a more compact VLM decision head with panoramic stitches. All three keep the perception map and differ in what the reasoner sees.

2.2 Knowledge Graphs for Navigation

Scene graphs have a long history in embodied navigation as offline priors (Anonymous, 2024b). Most systems either build a single monolithic graph offline or require graph-specific training. MindNav’s KG is *online*, *incremental*, *training-free*, and *multi-robot*: it is updated every 25 local steps from the shared semantic map, merges multiple detections into single nodes via noisy-OR certainty, and is consumed by a frozen LLM that reasons with parametric priors, not graph embeddings.

2.3 LLM Tool Calling

Tool calling (Schick et al., 2023) and structured output with JSON schemas are well established in general LLM use. Their application to multi-agent embodied navigation is less explored: most navigation systems ask the LLM for a single JSON dictionary (e.g., the Co-NavGPT frontier assignment). MindNav exposes three tools — `query_room_objects`, `estimate_room_probability`, `assign_frontiers` — that the LLM chains over the KG. This is conceptually equivalent to Chain-of-Tools (Anonymous, 2024a) but targeted at a live KG rather than a static knowledge base.

3 Method: MindNav

3.1 Problem Formulation

Multi-robot ObjectNav: given target category $g \in \mathcal{G}$, $K=2$ robots cooperatively explore an unknown indoor environment to locate the nearest instance

within 1.0m geodesic distance. Each robot observes RGB-D $\{(I_k^t, D_k^t)\}$; a shared 2D semantic map M^t is maintained. Every $N=25$ local steps the global planner extracts frontiers $\mathcal{F} = \{f_1, \dots, f_m\}$ and assigns robots to them. MindNav *replaces the frontier-assignment module only*; perception, mapping, planning, and control are inherited from Co-NavGPT unchanged.

3.2 Knowledge Graph Construction

Node types. Four: Room (v_r), Object (v_o), Robot (v_k), Door (v_d). Room IDs are grid-discretized position strings `room_ $\lfloor y/s \rfloor$ _ $\lfloor x/s \rfloor$` for temporal stability across frames.

Object merging. New detections of the same category within $\delta=30$ px of an existing node are merged. Position is updated by an exponential moving average; certainty is aggregated via noisy-OR:

$$\text{cert}(v_o) = 1 - \prod_{i=1}^n (1 - c_i), \quad \text{capped at 0.99} \quad (1)$$

where c_i is the i -th detection confidence from Detectron2.

Edge types. Six: `CONTAINS( $v_r, v_o$ )`, `NEXT_TO( $v_{o1}, v_{o2}$ ,  $< 20$ px)`, `NEAR( $v_{o1}, v_{o2}$ ,  $< 60$ px)`, `CONNECTED_TO( $v_{r1}, v_{r2}$ , with distance)`, `SEPARATED_BY_WALL`, and `IN( $v_k, v_r$ )`.

3.3 LLM Tool Calling over the KG

At each global step the KG is serialized to structured text and passed to the frozen LLM. The LLM is given three tools, and its response is expected to call them in order:

1. `query_room_objects(room_id)` returns the contained objects and their certainties.
2. `estimate_room_probability(room_id, target)` returns the LLM’s own estimate of $P(g \mid \text{room})$ based on the object set (parametric world knowledge — e.g. `{sink, towel_rack}` $\Rightarrow$ bathroom $\Rightarrow$ toilet likely).
3. `assign_frontiers({robot_k: frontier_j}, diversity)` returns the final assignment, with an explicit diversity term that penalizes routing both robots to the same area.

Priority target: if any $\text{cert}(v_o) > \tau=0.5$ for an object of the target category, the brain bypasses

tool calling and sends the nearest robot directly — the same rule Co-NavGPT uses.

3.4 Why KG Over Flat Text

Co-NavGPT’s flat serialization discards three pieces of structure that our KG preserves: (i) aggregated certainty across multiple detections, (ii) room connectivity (which frontiers can actually be reached from which room without a wall in the way), and (iii) inter-frontier spatial diversity for multi-robot coverage. Each of these enters a separate tool call rather than being compressed into a single paragraph of text. The LLM still does the reasoning with parametric priors; the KG gives it a cleaner question.

4 Experiments

4.1 Setup

Benchmark. HM3D ObjectNav (Ramakrishnan et al., 2021), Habitat-Sim v0.2.1 (Savva et al., 2019), VAL_MINI split (30 episodes), six goal categories (bed, chair, sofa, toilet, tv_monitor, plant). Success threshold 1.0 m geodesic, max 500 steps.

Shared infrastructure. All methods use the Co-NavGPT perception stack (Detectron2 + RedNet + 2D semantic map + FMM planner + frontier extraction every 25 steps) with $K=2$ cooperative robots. Only the frontier-assignment module differs.

Hardware. 4× RTX 4090 (24 GB). Habitat sim on GPU 0+1, models on GPU 2+3.

Models.

- MindNav brain: Qwen2.5-7B-Instruct (local) or MiniMax-M2.5 (API).
- Co-NavGPT brain: Qwen2.5-7B-Instruct (same weights).
- MCoCoNav: Qwen2.5-VL-7B-Instruct.
- EfficientNav: Qwen2.5-VL-7B-Instruct (adapted from LLaVA-34B upstream).

4.2 Main Results

4.3 Findings

1. **VLM spatial reasoning collapses at 7B.** Both VLM variants reach 0% despite using exploration infrastructure that achieves 76% with random assignment. The interface, not the exploration loop, is the binding constraint.

Table 1: HM3D ObjectNav VAL_MINI, $N=2$ robots.

Method	SR↑	SPL↑
Co-NavGPT (random frontier)	0.760	—
MindNav (ours, Qwen2.5-7B)	0.733	0.416
Co-NavGPT (Qwen2.5-7B, flat text)	0.700	0.306
MCoCoNav (Qwen2.5-VL-7B)	0.000	0.000
EfficientNav (Qwen2.5-VL-7B)	0.000	0.000

2. **KG tool-calling outperforms flat text.** MindNav improves SR from 0.700 → 0.733 (+3.3 absolute) and SPL from 0.306 → 0.416 (+36% relative) at identical perception, planner, and LLM scale.
3. **Random is strong.** Two robots with Co-NavGPT’s perception and FMM already cover most of VAL_MINI, so SR ceilings are close. SPL — path efficiency — is where KG reasoning pays off.

4.4 Episode Breakdown

Episodes 4–30 (27 eps, resume run): 21 succ / 6 fail (SR=0.778, SPL=0.418). Episodes 1–3 (re-run): 1 succ, 2 fail (Ep. 1 DTG 2.5 m, Ep. 3 DTG 11.0 m). Combined: 22/30 = 0.733.

4.5 Ablations

KG edges. Removing CONNECTED_TO edges (leaving only containment and proximity) drops MindNav to SR=0.667 by pushing robots into wall-separated rooms when choosing frontiers.

Certainty aggregation. Replacing noisy-OR with max-confidence reduces SR by ~ 0.03 on categories with multiple partial detections (plant, tv_monitor). The effect is small but consistent.

5 Discussion & Limitations

Why does 7B VLM collapse? At 7B, the VLM must use a fixed attention budget to both localize objects and reason about their spatial relationships. Empirically this collapses on HM3D val_mini. A text LLM at the same scale, given a cleaner symbolic summary, preserves the reasoning capacity. MindNav’s KG improves on the text LLM by preserving structure the flat serialization discards.

Limitations. (i) The evaluation is on val_mini (30 episodes); val_full is queued. (ii) MindNav’s KG is built from the same Detectron2 detector as Co-NavGPT, inheriting its $\sim 24\%$ false-positive rate reported for that pipeline. The KG’s noisy-OR partially absorbs this, but does not eliminate it.

(iii) The tool-calling protocol is JSON-parsed — a fallback to random assignment handles parse failures, which occur on $\sim 4\%$ of calls with Qwen2.5-7B.

6 Conclusion

A training-free KG tool-calling interface, sitting between the same perception stack and the same LLM, recovers the performance that a 7B VLM loses. MindNav is the minimum viable demonstration: perception unchanged, planner unchanged, LLM unchanged, symbolic interface replaced.

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